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PALAEOMAGNETISM OF THE TORORO RING COMPLEX, S.E. UGANDA

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Palaeomagnetic measurements on the pre-Miocene carbonatite volcanics of Tororo, S.E. Uganda, have yielded a pole at 75.8°N, 195.5°E with A_{95} = 9.4°. Along with the Tertiary poles from East African rift systems, the Eocene–Oligocene pole from Ethiopia and the mean Mesozoic pole from the rest of Africa, a polar wander path for Africa from *Mesozoic to present* is suggested.

1. Introduction

Palaeomagnetic results from the Miocene to recent volcanics of the East African rift system have been reported by several authors [1–3]. The pole position for the Ethiopian flood basalts [4] considered to be Eocene-Oligocene in age shows a 9° deviation from the present pole as compared to the small 5° deviation of the Miocene pole. The next oldest known pole for Africa is the one obtained from the Lupata series volcanics [5] with an age of ~110 my [6]. For the period 200 my–110 my, eight other poles are available and because of their close grouping, Briden [7] concluded that the African continent was stationary during this period. The mean of these nine poles (64.5°N, 261°E with A_{95} = 5°) spread through the period Cretaceous to Triassic was therefore taken as the mean Mesozoic pole by McElhinny et al. [6]. Pole positions for rocks of intermediate age are lacking because of the unavailability of such rocks in East Africa. The result obtained from the alkaline complexes of Tororo and presented here helps to fill a gap in the polar wander path for Africa.

2. Geology and age

The alkaline complexes of S.E. Uganda lie on a line which runs a little east of north from Sukulu through Tororo, Bukusu, Sekululu to Budeda. They are broadly similar, with a central carbonatite core surrounded by a ring of syenite and ijolite. Carbonatite forms the main mass of the prominent outcrops at Tororo hill, from where the samples were collected.

No radiometric date is available for these formations. They are presumed to be "pre-Miocene, post Karroo (probably Cretaceous)" in age [8]. The Carbonatite volcanoes of the Kenya–Uganda border have given a consistent radiometric age ranging from 18–20 my [9]. However, based on stratigraphy of the area, Davies [8] considers the alkaline complexes of Uganda to be older than those of the Kenya–Uganda border and according to him the close proximity of the two "suggests the maintenance of a magma-reservoir of consistent qualities in the subcrustal regions over a long period of geological time". Palaeomagnetic evidence presented in this paper supports the view that the Uganda formations are pre-Miocene.

3. Sampling and measurement

Suitable exposures were located only in the lime kiln quarries in Tororo hill. Twenty three cores of 2.2 cm diameter were collected from three sites in this area using a portable drill of the type described by Doell and Cox [10]. Site 3 in the uppermost quarry was about 210 m higher than site 2 in the lowermost quarry. Site 1 was about 150 m below

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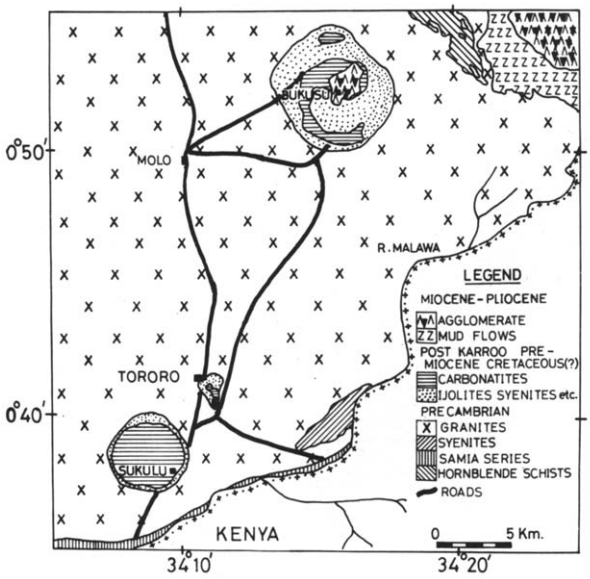


Fig. 1. Geological sketch map of South East Uganda.

site 3. Fig. 1 is the geological sketch map of the region.

A total of 34 specimens of length 2.3 cm each were cut from the above samples and their NRM measured under an astatic magnetometer with maximum sensitivity 1.3×10^{-5} emu cm^{-3}. The specimens were subjected to stepwise A.F. demagnetization in peak fields of up to 250 Oe. Because of considerable fall in inten-

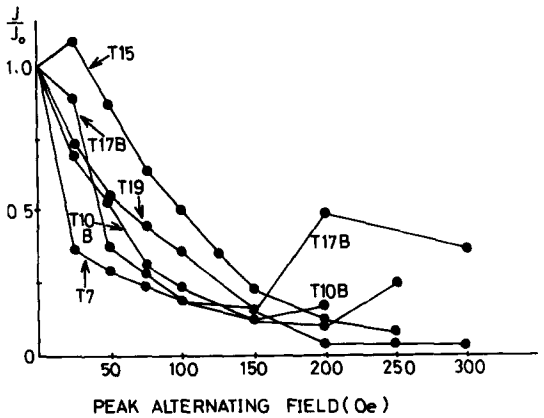


Fig. 2. A.F. demagnetization curves of typical specimens from each site.

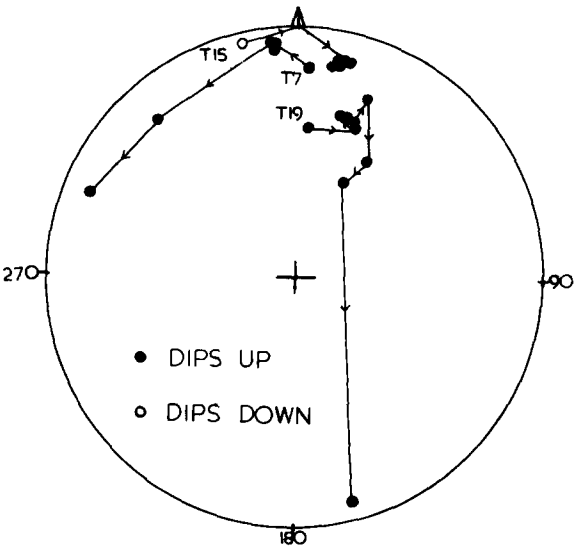


Fig. 3. Changes in direction of typical specimens during demagnetization in steps of 25 Oe.

TABLE 1

Site no.	Specimen no.	NRM direction		Cleaned direction	
		D	I	*D*	I
1	T 2	325	−56	343	− 8
	T 5	38	−50	8	−17
	T 6	328	+ 8	11	−45
	T 7	2	−18	354	−10
2	T 9A	38	−20	3	−37
	T 9B	33	−28	9	−40
	T 9C	6	− 9	358	−11
	T 10A	59	−28	9	−27
	T 10B	354	−40	12	−28
	T 11A	338	−22	24	−10
	T 11B	69	−23	11	−29
	T 12	306	−41	10	−26
3	T 13	103	− 4	10	−34
	T 14	36	−32	7	−23
	T 15	346	+ 5	9	−17
	T 17A	348	−53	25	−35
	T 17B	342	+48	346	−31
	T 18	354	−34	14	−38
	T 19	4	−41	14	−34
	T 20A	309	−16	10	−14
	T 20B	343	−46	353	−35
	T 21	357	−21	11	−22
	T 22A	21	−16	21	−16
	T 22B	357	−28	14	−35
	T 23B	340	−13	340	−13
	T 23C	352	−12	357	−11

TABLE 2
The site location of Tororo carbonatite volcanics is lat. 0.7°N, long. 34.2°E

Site no.	No. of samples	No. of specimens	Site mean direction (after cleaning)		k (for specimens)	α_{95}°	Palaeomagnetic lat.°N	North pole long.°E
			D^0	I^0				
1	4	4	358	−20.2	15.2	24.3	78.7	224.3
2	4	8	9.6	−26.2	40.1	8.9	72.6	181.3
3	10	14	6.3	−26.1	28.3	7.6	74.2	191.1
Overall mean for the three sites			4.5	−24.2	157	9.9	75.8	195.5 with A_{95}=9.4° and K=172.2

k, K, Fisher's precision parameter.
α_{95}, A_{95}, radius of 95% circle of confidence.

sity during demagnetization, a commercial version of the low frequency spinner magnetometer of the Foster type [11] was used for measurement of the demagnetized specimens.

The NRM intensity varied widely from specimen to specimen ($43.3 \times 10^{-3} - 28.9 \times 10^{-6}$ emu cc^{-1}) and the wide variation of intensity between specimens from the same sample suggests that the magnetic material is very inhomogeneously dispersed in the body of the rock. In most specimens soft secondary components of magnetization were removed in peak fields of about 75 Oe. However, excepting a small number of specimens, the rest behaved erratically in fields above 150 Oe both as regards direction as well as intensity. The behaviour of typical specimens from each site is shown in figs. 2 and 3.

Specimens from samples T1, T3, T8 and T16 behaved erratically during demagnetization and no stable direction could be obtained from them. Hence these samples were rejected as unsuitable for palaeomagnetic purposes. Fig. 4 gives the directions obtained from the rest before and after demagnetization.

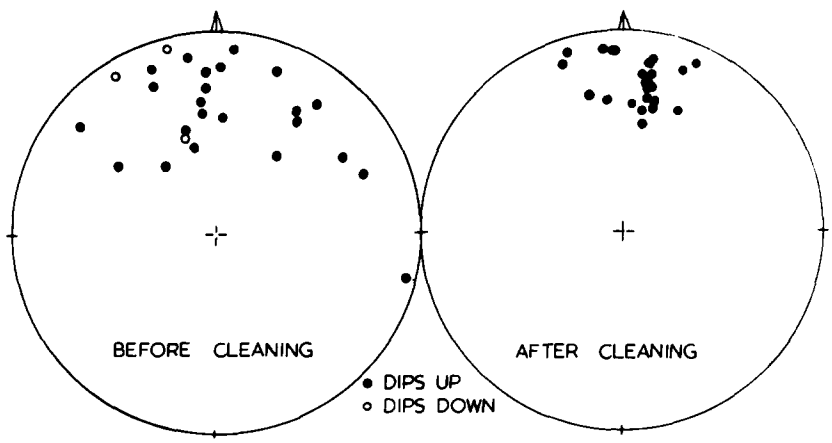


Fig. 4. Directions of magnetization of the 26 specimens before and after cleaning.

4. Results and discussion

Table 1 gives the specimen directions before and after demagnetization. In the analysis, site means were calculated giving unit weight to specimens. The pole for each site was then calculated from the site mean direction and the overall mean pole for the formation deduced from the three site poles using the well known Fisheries statistics. These are presented in table 2.

The youngest pole from East Africa is shown as R_1 in fig. 5. This pole (88°N, 144°E with A_{95} = 3°) is the one obtained from the rift valley volcanics of the age group 0–2.5 my [2]. The next oldest pole R_2 (88°N, 153.5°E and A_{95} = 3°) is also from the rift volcanics, but of age 2.5–7 my [2]. These two poles are not significantly different from the present geographical pole and their circles of confidence enclose the present pole. R_3 (85°N, 140°E, A_{95} = 3.5°) and V (85°N, 136.5°E, A_{95} = 3°) are poles obtained from detailed palaeomagnetic study of the rift volcanics [2] of age 7.0–14 my and of the Miocene volcanics of Cape Verde Islands [12]. The remarkable closeness of these two poles supports the view that Cape Verde Islands are part of the African plate.

The Miocene pole T (85°N, 163.5°E, A_{95} = 2.5°) from the Turkana lavas [1] is the oldest of the East African Tertiary poles. Combining the above poles with the Ethiopian pole E (81°N, 168°E, A_{95} = 4.5°) deduced by Brock et al. [4] from the Eocene–Oligocene Ethiopian flood basalts, enables one to construct a well defined Cenozoic polar wander path for Africa. The Australian polar wandering during the Cenozoic has been compared by McElhinny et al. [13] with the African polar wander curve and the close similarity between the two enabled him to suggest the existence of a common polar wandering component. Further, the agreement of the African polar wander path with that of Europe and Northern Asia on the one hand and with the polar wander path for Antarctica on the other, rules out Girdler's [14] suggestion of rotation of Africa during the *Upper* Tertiary and suggests that the excursions of the African pole by 9° in the Eocene and by 5° in the Miocene in the direction of the 160°E meridian are solely due to polar wander.

The pole from the present study (75.8°N, 195.5°E, A_{95} = 9.4°) is shown as *TU* in fig. 5. Its position between the mean Mesozoic pole M and the Eocene–Oligocene Ethiopian pole E is consistent with the age "pre-Miocene, probably Cretaceous" suggested by Davies [8]. The pole thus helps to fill a gap in the polar wander curve for Africa, even though its precise age is not known. Fig. 5 shows the suggested polar wander path since the Mesozoic, taking into account all the available Tertiary poles.

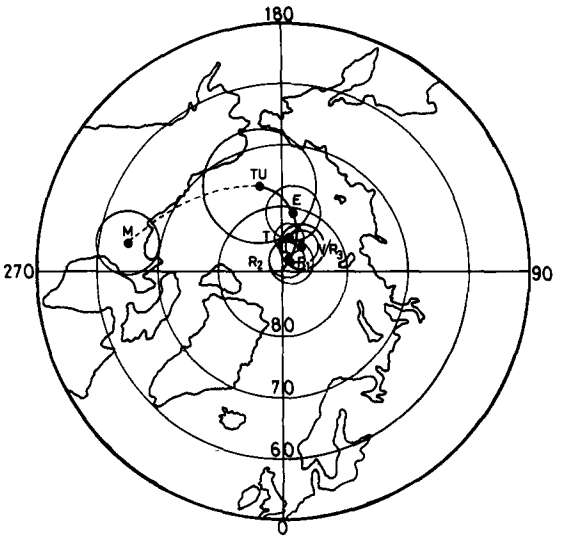


Fig. 5. Polar wander path for Africa from *Mesozoic to present*. R_1, R_2, R_3 are poles from rift valley volcanics of age groups 0–2.5, 2.5–7.0 and 7.0–14 my respectively. V and T are Miocene poles from Cape Verde Islands and Turkana. E – Eocene–Oligocene pole from Ethiopia. M – Mean Mesozoic pole. *TU* – pole obtained from the present study.

The 14.2° deviation of the pole *TU* from the present pole may be due to polar wander as well as continental drift, but it is not possible to separate one from the other until more results from rocks intermediate in age between Eocene and Mesozoic are available.

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